

Basic Prob. Facts LOTUS: $E[g(X)] = \sum_x g(x) P(X=x)$ Tower: $E[E[X|Y]] = E[X]$
 Expectation: $E[X] = \sum_x x P(X=x)$ Conditional Prob: $P(X=x|Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$ $\sum_x P(X=x|Y=y) = 1$
 Variance: $Var(X) = E[(X-E[X])^2] = E[X^2] - E[X]^2$ Total exp: $E[X] = \sum_y P(Y=y) E[X|Y=y]$
 $Var(\sum_i X_i) = \sum_i \sum_j Cov(X_i, X_j) = \sum_i Var(X_i) + 2 \sum_{i < j} Cov(X_i, X_j)$ MGF: $M_X(s) = E[e^{sX}] = \sum_x e^{sx} P(X=x)$
 Covariance: $Cov(X, Y) = E[(X-E[X])(Y-E[Y])] = E[XY] - E[X]E[Y]$ $\frac{d^n}{ds^n} M_X(s) \Big|_{s=0} = E[X^n]$
 $Cov(X, aY+b) = a Cov(X, Y)$ $Cov(X, Y+Z) = Cov(X, Y) + Cov(X, Z)$ $E[e^{sX}] = E\left[\sum_x E[X^k] \frac{s^k}{k!}\right]$
Concentration Inequalities
 Markov: $P(X \geq a) \leq \frac{E[X]}{a}$
 Chebyshev: $P(|X-E[X]| \geq \epsilon) \leq \frac{Var(X)}{\epsilon^2}$ in probability: $P(|X_n - X_\infty| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$
 Bounded Convergence Thm: $X_n \xrightarrow{a.s.} X_\infty$ as $n \rightarrow \infty$ and $|X_n| \leq C$ for $n \geq 0$ and $\omega \in \Omega$. Then $E[X_n] \rightarrow E[X_\infty]$ as $n \rightarrow \infty$
 Dominated Convergence Thm: $X_n \xrightarrow{a.s.} X_\infty$ as $n \rightarrow \infty$ and $\exists$ r.v. Y s.t. $E[Y] < \infty$ and $|X_n(\omega)| \leq Y(\omega)$ for $n \geq 0, \omega \in \Omega$. Then $E[X_n] \rightarrow E[X_\infty]$ as $n \rightarrow \infty$
 Monotone Convergence Thm: $(X_n, n \geq 0)$ nondecreasing, $X_n(\omega) \leq X_{n+1}(\omega)$, $n \geq 0, \omega \in \Omega$. Then $X_\infty = \lim_{n \rightarrow \infty} X_n$ exists and $E[X_n] \uparrow E[X_\infty]$
 Fatou's Lemma: $(X_n, n \geq 0)$ nonnegative means $E \liminf X_n \leq \liminf E[X_n]$
 CLT: $(X_n)_{n=1}^\infty$ iid with mean μ , variance σ^2 . $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{indist} N(0,1)$ as $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \Rightarrow N(0,1)$
 SLLN: $(X_n)_{n=1}^\infty$ iid, X_i 's bounded. $\bar{X}_n \xrightarrow{a.s.} E[X_1]$
 WLLN: iid $Z_1, \dots, Z_n$. $\bar{Z}_n \xrightarrow{P} E[Z_1]$
 Ergodic Thm: $(Z_n, n \geq 0)$ stationary, $E[Z_0] < \infty$. Then $\exists W$ where $E[W] = E[Z_0]$ and $\bar{Z}_n \xrightarrow{a.s.} W$ as $n \rightarrow \infty$. If W deterministic, Z_n is ergodic.
Markov Chains
 Markov Property: $P(X_{n+1} \in B | X_0, \dots, X_n) = P(X_{n+1} \in B | X_n) = P(X_1 \in B | X_0) = P(X_n, X_{n+1})$
 Stationary distribution: chain will have the same dist for all time: $\pi P = \pi$ (solve s.t. $\pi e = 1$)
 $P(X, B) = \sum_{b \in B} P(X, b)$ or $P(X, B) = \int P(X, b) db$ state is transient iff $\sum_{n=0}^\infty P^n(x, y) < \infty$
 Hitting time $T = \inf \{n \geq 0 : X_n \in C\}$ (first time we hit C) Recurrent otherwise
 n-step transition probabilities: $\gcd \{n \geq 1 : P^n(x, x) > 0\} = p \Rightarrow x \in S$ is periodic of period p . If the period is 1, the state is aperiodic.
 For an irreducible positive recurrent aperiodic MC, $P^n(x, y) \rightarrow \pi(y)$ as $n \rightarrow \infty$ where $\pi(y)$ is the stationary dist at state y .
 When you condition an MC on the outcome of an event, it's an MC.
 Reversing an MC: $X \rightarrow X'$ is also an MC with transition matrix $\tilde{P}(x, y) = \frac{\pi(y) P(y, x)}{\pi(x)}$
 Global balance: $\pi = \pi P$ Detailed balance: $\pi(x) P(x, y) = \pi(y) P(y, x) \forall x, y \in S$

Asymptotic loss of memory

$$p \geq \delta \cdot 1$$

$$p = \delta \cdot 1 + (1-\delta)Q, \text{ where } Q = \frac{p - \delta \cdot 1}{1-\delta}$$

$$p^2 = \delta \cdot 1 + (1-\delta)\delta \cdot 1Q + (1-\delta)^2Q^2$$

$$\vdots$$

$$p^n = \delta \cdot 1 + \delta(1-\delta)1Q + \dots + (1-\delta)^nQ^n$$

$$p^n = \sum_{j=0}^{\infty} \delta(1-\delta)^j 1Q^j = \pi \text{ (stochastic, identical rows)}$$

i.e. $p^n(x,y) \rightarrow \pi(y)$ for irreducible recurrent aperiodic MC

$$\|p^n - \pi\| \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\text{where } \| \cdot \| = \sup_{\|f\|_p=1} \{ \|Af\|_p \}$$

Martingale - expected value of the

next observation, given all prior observations, is equal to the most recent value. formally, a sequence $(M_n, n \geq 0)$ is a Martingale (MG) adapted to $(\mathcal{Z}_n, n \geq 0)$ if $E[M_n] < \infty$ for $n \geq 0$

and $E[M_{n+1} | \mathcal{Z}_0, \dots, \mathcal{Z}_n] = M_n$ a.s. $\geq M_n$ is a submartingale and $\leq M_n$ is a supermartingale

$E[M_0] = E[M_n]$. The differences D_i, D_j are uncorrelated. "Adapted to": a sequence $(Y_n, n \geq 0)$ is adapted to $(\mathcal{Z}_n, n \geq 0)$ if for each $n \geq 0$, $\exists$ deterministic function f_n s.t. $Y_n = f_n(\mathcal{Z}_0, \dots, \mathcal{Z}_n)$.

Optional Sampling - If $(Y_n, n \geq 0)$ adapted to $(\mathcal{Z}_n, n \geq 0)$ then T is a stopping time adapted to $(\mathcal{Z}_n, n \geq 0)$. Also, $E[M_{\min}(T, m)] = E[M_0] \forall m \geq 1$.

Practice Exam Q1: $T = \inf\{n \geq 0: X_n \in C^c\}$, $C^c \in \mathcal{S}$, $u^* = E[x \cdot z^T g(X_T)]$

Apply FTA: $u(x) = z E_x[g(X_T)] I(X_1 \in C^c) + z \sum P(x,y) u(y)$ $u = f + Bu$

Vectors: $f(x) = z E_x[g(X_T)] I(X_1 \in C^c)$ $B = P(x,y)$

for $x \in C$.

$$u^*(x) = E_x[z^T g(X_T)]$$

$$= E_x[z^T g(X_T) I(T < \infty)]$$

$$= \sum_{n=1}^{\infty} z^n E_x[g(X_n) I(T=n)]$$

$$= \sum_{n=1}^{\infty} z^n \sum_{x_1, \dots, x_n} P(x_1=x_1, \dots, X_n=x_n) f(x_n)$$

$$= \sum_{n=1}^{\infty} z^n \sum_{x_1, \dots, x_n} P(x_1, x_2) \dots B(x_n, x_{n-1}) \leq P(x_{n-1}, x_n) f(x_n)$$

$$= \sum_{n=1}^{\infty} z^n \sum_{x_1, \dots, x_n} B^{n-1}(x, x_{n-1}) \leq z P(x_{n-1}, y) f(y)$$

$$= \sum_{n=0}^{\infty} z^n \sum_{x_1, \dots, x_n} B^n(x, x_n) f(x_n)$$

$$= \sum_{n=0}^{\infty} (z^n B^n f)(x); \text{ i.e. } u^* = \sum_{n=0}^{\infty} (z^n B^n f)(x)$$

First Step Analysis: example for calculating $T = \inf\{n \geq 0: X_n \in C^c\}$

$$u^* = P_x(X_1 \in A), T < \infty \text{ for } A \in C^c$$

$$u^*(x) = \sum P(x,y) u^*(y) \quad x \in C$$

where $u^*(z) = 1$ if $z \in A$ and $u^*(z) = 0$ if $z \in C^c - A$

(can also be expressed without boundary conditions: $u = f + Bu$)

FSA may not have a $f(x) = P_x(X_1 \in A)$

unique solution, but the $x \in C$ meaningful solution u^* is given by $u^* = \sum_{k=0}^{\infty} B^k f$

Results from FSA - the following are equivalent:

① $G_n \rightarrow 0$ as $n \rightarrow \infty$ ② $\sum_{n=0}^{\infty} G^n < \infty$ ③ $(I - G)^{-1}$ exists and is nonnegative

Some MR Results ④ $(Y_n, n \geq 0)$ stationary $E[Y_n] < \infty$ $\text{Cov}(Y_0, Y_n) \rightarrow 0$ as $n \rightarrow \infty$. Then $\bar{Y}_n = \frac{1}{n} \sum_{i=0}^{n-1} Y_i \rightarrow E[Y_0]$

⑤ $(Y_n, n \geq 0)$ stationary $E[Y_0^2] < \infty$ $\text{Var}(Y_0) + 2 \sum_{i=1}^{\infty} \text{Cov}(Y_0, Y_i) < \infty$

Then $\bar{Y}_n \xrightarrow{a.s.} E[Y_0]$

sequence $(M_n, n \geq 0)$ is a Martingale (MG) adapted to $(\mathcal{Z}_n, n \geq 0)$ if $E[M_n] < \infty$ for $n \geq 0$

and $E[M_{n+1} | \mathcal{Z}_0, \dots, \mathcal{Z}_n] = M_n$ a.s. $\geq M_n$ is a submartingale and $\leq M_n$ is a supermartingale

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for $x \in C$.

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$$= \sum_{n=1}^{\infty} z^n \sum_{x_1, \dots, x_n} P(x_1=x_1, \dots, X_n=x_n) f(x_n)$$

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$$= \sum_{n=1}^{\infty} z^n \sum_{x_1, \dots, x_n} B^{n-1}(x, x_{n-1}) \leq z P(x_{n-1}, y) f(y)$$

$$= \sum_{n=0}^{\infty} z^n \sum_{x_1, \dots, x_n} B^n(x, x_n) f(x_n)$$

$$= \sum_{n=0}^{\infty} (z^n B^n f)(x); \text{ i.e. } u^* = \sum_{n=0}^{\infty} (z^n B^n f)(x)$$

so that $\text{Cov}_{\pi}(f(X_0), f(X_n)) = O((1-\delta)^n)$ Ths.

$\frac{1}{n} \sum_{j=0}^{n-1} f(X_j) \rightarrow \int_{\mathcal{S}} f(x) \pi(dx)$ P π a.s.

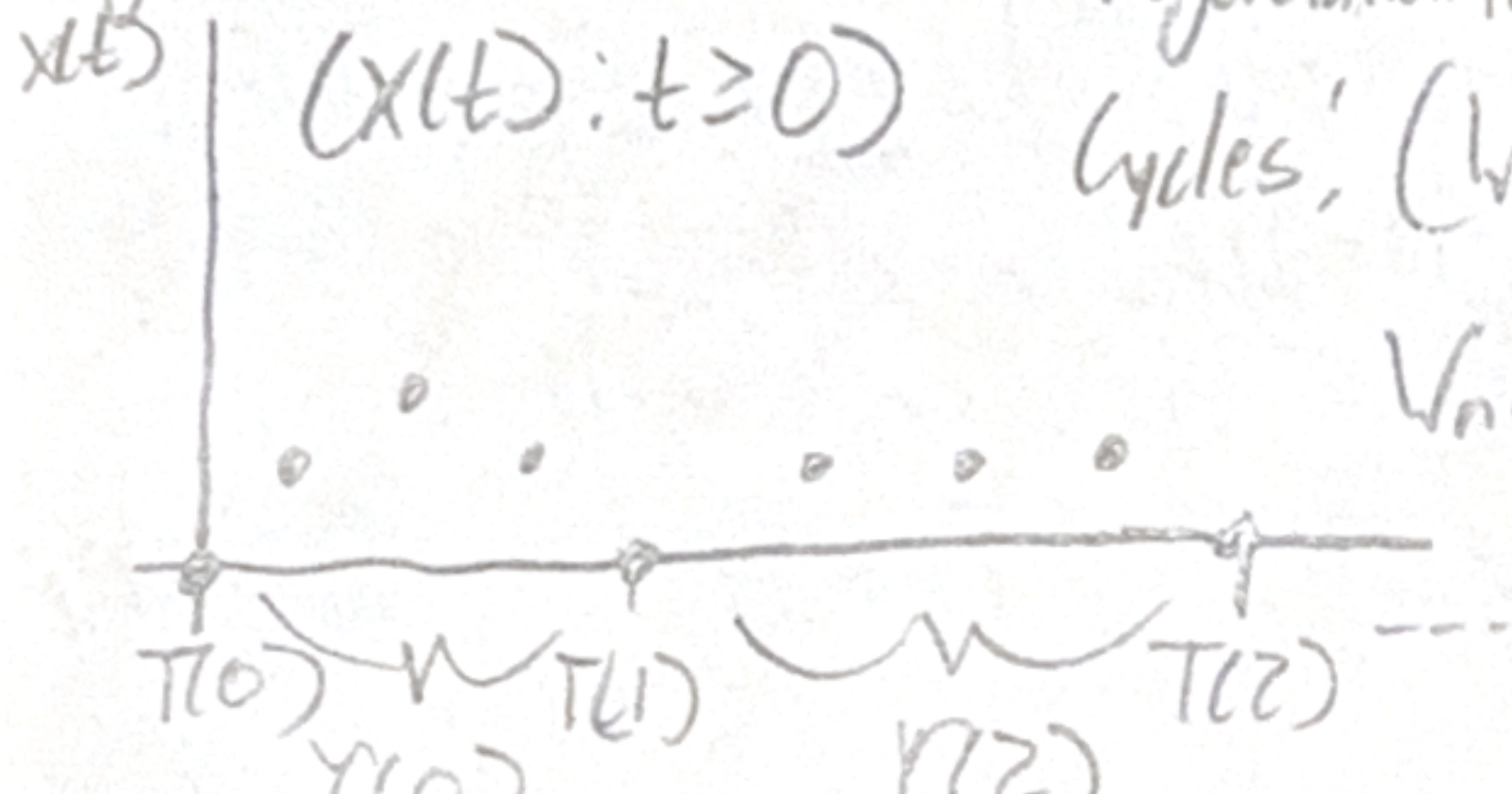
Martingale SLN

$\frac{1}{n} \sum_{i=1}^n P_i \rightarrow 0$ a.s. (if $\sum P_i < \infty$)

Potential Eqn
 $(P-I)f = h$

potential, cost or accumulated deviation from the mean

Regenerative Processes



Strong Law for Regenerative Processes:

If $\int_0^{T(1)} f(X(s)) ds < \infty$ a.s.
 $E[\int_0^{T(1)} f(X(s)) ds]$ and $E[Y_1]$ are not simultaneously infinite
 then $\frac{1}{t} \int_0^t f(X(s)) ds \rightarrow \frac{E[Y_1(f)]}{E[Y_1]}$ a.s.
 where $Y_1(f) = \int_0^{T(1)} f(X(s)) ds$

Harris Chains (recurrence without strong $\pi = \pi P$)
 MC $(X_n: n \geq 0)$ is Harris recurrent if $\exists A \subseteq S, \lambda > 0, m \geq 1$ and a distribution ψ on S s.t.

- i) $P_x(T_A < \infty) = 1, x \in S$, where $T_A = \inf\{n \geq 0: X_n \in A\}$
- ii) $P_x(X_m \in \cdot) \geq \lambda \psi(\cdot), x \in A$

If X is discrete, X is Harris recurrent iff $\exists z \in S$ s.t. $P_x(\nu(z) < \infty) = 1$ for $x \in S$, where $\nu(z) = \inf\{n \geq 1: X_n = z\}$

For Harris recurrent MC and non-negative f

$\frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \rightarrow \int \pi(y) f(y) dy$ a.s. as $n \rightarrow \infty$

Martingale CLT

$(M_n: n \geq 0)$ square integrable

$\frac{1}{n} [M](n) \rightarrow \sigma^2$ as $n \rightarrow \infty$

$[M](n) = M^2(0) + \sum_{i=1}^n P_i^2$

$\frac{1}{n} \langle M \rangle(n) \rightarrow \sigma^2$ as $n \rightarrow \infty$

$\langle M \rangle(n) = \sum_{i=1}^n E[P_i^2 | \mathcal{F}_{i-1}]$

Then $\frac{1}{\sqrt{n}} \sum_{i=1}^n P_i \Rightarrow N(0,1)$

Martingale Convergence Thm

$(M_n: n \geq 0)$ adapted to $(\mathcal{F}_n: n \geq 0)$
 If $\sup_{n \geq 0} E[M_n^2] < \infty, \exists$ square integrable r.v. M_∞ s.t.
 $M_n \rightarrow M_\infty$ a.s. as $n \rightarrow \infty$

Supermartingale Convergence Thm

$(M_n: n \geq 0)$ super MC, nonnegative

Then, $\exists$ finite-valued r.v. M_∞ s.t. $M_n \rightarrow M_\infty$ a.s. as $n \rightarrow \infty$

And $\sqrt{n}(\hat{\theta}_n - \theta^*) H_n = \frac{-M_n}{n} \Rightarrow N(0,1)$

$S = E[P^2]$

Data-Driven Markov Modeling

Goal: learn parameters θ to learn the transition probabilities, i.e.

$\hat{\theta} = \arg \max_{\theta} d(\theta) = \sum_{i=1}^n \log P(\theta, X_i, X_{i+1})$

$\Rightarrow M_n = \sum_{i=0}^{n-1} \frac{\nabla P(\theta^*, X_i, X_{i+1})}{P(\theta^*, X_i, X_{i+1})}$ is M_0

$\sqrt{n}(\hat{\theta}_n - \theta^*) H_n = \frac{-M_n}{n} \Rightarrow N(0,1)$

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Regeneration times: $(T(n): n \geq 0)$

Cycles: $(Y(n): n \geq 0)$

$V_n(t) = \sum_{i=0}^{n-1} X(T(i-1) + t) \mathbb{1}_{0 \leq t < T_n}$

$\Delta t \geq T_n$

CLT for Regenerative Processes

$X = (X(t): t \geq 0)$ S-valued regenerative process with regeneration times $(T(n): n \geq 0)$

Sps. $f: S \rightarrow \mathbb{R}$ satisfies $Y_0(f) < \infty$ a.s. and

$E[Y_1(f)^2] < \infty$. Then

$\sqrt{t} \left(\frac{1}{t} \int_0^t f(X(s)) ds - \alpha \right) \Rightarrow N(0,1)$

where $\alpha = \frac{E[Y_1(f)]}{E[Y_1]}, \sigma^2 = \frac{E[Z_1^2]}{E[Y_1]}$

and $Z_1 = Y_1(f) - \alpha Y_1$

Thm. 11.2.1. Sps. $\exists$ AES $\lambda > 0, n \rightarrow \infty$

$m \geq 1$, a distribution $\psi, \epsilon > 0$, and $g: S \rightarrow [0, \infty)$ s.t.

i) $E_x[g(X_1)] \leq g(x) - \epsilon, x \in A^c$

ii) $\sup_{x \in A} E_x[g(X_1)] < \infty$

iii) $P_x(X_m \in \cdot) \geq \lambda \psi(\cdot), x \in A$

then, we have a Harris chain.

Also, g is the Lyapunov function.

Common candidates for g

include $\|x\|^p, \exp(a\|x\|^p)$, and

$(\log(1 + \|x\|))^p$

Maximal Inequality

$P(\max_{1 \leq i \leq n} M_i > x) \leq \frac{E[|M_n|]}{x}$

X is regenerative if

$\bullet V_0, W_1, \dots$ are independent

$\bullet W_1, W_2, \dots$ are iid

When $T(0) = 0$, this is a renewal counting process.

$\frac{M(t)}{t} \rightarrow \frac{1}{E[Y_1]}$ a.s. as $t \rightarrow \infty$

Elementary renewal thm:

$\frac{E[M(t)]}{t} \rightarrow \frac{1}{E[Y_1]}$ as $t \rightarrow \infty$

Recurrence A recurrent state $z \in S$ is said to be positive

recurrent if $E_z[\nu(z)] < \infty$

and null recurrent otherwise

Let $X = (X_n: n \geq 0)$ be MC.

If X is null recurrent and

$f: S \rightarrow \mathbb{R}_+$ s.t. $E[Y_1(f)] < \infty$,

then $\frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \rightarrow 0$ a.s. as $n \rightarrow \infty$

Also, if $E[Y_1(f)] < \infty$, then

$\frac{1}{n} \sum_{i=0}^{n-1} E_x[f(X_i)] \rightarrow 0$

If X is positive recurrent iff

$\exists$ a prob. soln. to $\pi = \pi P$

If X is positive recurrent, then

for each $f: S \rightarrow \mathbb{R}_+, \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \rightarrow \int \pi(y) f(y) dy$ a.s. as $n \rightarrow \infty$

and $\frac{1}{n} \sum_{i=0}^{n-1} E_x[f(X_i)] \rightarrow \int \pi(y) f(y) dy$ as $n \rightarrow \infty$

Also, $\pi(z) = \frac{1}{E_z[\nu(z)]}$

Contrasting MC

$X = (X_n: n \geq 0)$ is a Markov process with iid

$X_n = (a_n, a_{n+1}, \dots, a_{n+m})$ ϕ_i 's iid

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$X_n = (a_n, a_{n+1}, \dots, a_{n+m})$ ϕ_i 's iid

$P(\max_{1 \leq i \leq n} M_i > x) \leq \frac{E[|M_n|]}{x}$

$E[\nu(z)] \leq C \phi(x, y) \leq 1$

Stochastic Control (formulation below)

S : state space
 $A(x)$: set of actions available in $x \in S$
 $P_a(x, dy)$: probability of going from x to dy under choice of action $a \in A(x)$
 $r(x, a)$: reward earned by using action $a \in A(x)$ in state x

$$V^*(x) = \max_{a \in A(x)} \left[r(x, a) + \int_S P_a(x, dy) V^*(y) \right]$$

Monotone MC $S = \mathbb{R}^+$
 $P_1(x) \leq P_2(y) \quad x \leq y$
 $X_n = (p_1, p_2, \dots, p_n)(x)$
 $W_n = M_n = \max_{0 \leq k \leq n} S_k \quad P(V \leq x+z)$
 $X_n \Rightarrow \tilde{X}_\infty$ provided $\tilde{X}_\infty$ finite-valued a.s.

Practice Final

1. $T(n) = \tau_1 + \tau_2 + \dots + \tau_n$ is the time when the n th component is replaced. If $N(m) = \max \{n \geq 0 : T(n) \leq m\}$ then $X_m = m - T(N(m))$ is the age.
 X_m is an MC: $P(X_{m+1} = 0 | X_0 = x_m) = \frac{P(X_{m+1})}{\sum_{j > x_m} P(j)}$
 $P(X_{m+1} = 0 | X_0 = x_m) = \frac{\sum_{j > x_m} P(j)}{\sum_{k > x_m} P(k)}$

2. $X_{n+1} = (X_n + Z_{n+1})^{\delta}$, $\delta \in (0, 1)$. $E[Z_i] < \infty$, show Harris chain.
 Choose $g(x) = x^{1/\delta}$
 i) $E_x[g(X)] \leq g(x) - \epsilon$, $x \in A^c$? $\checkmark$
 $E[X_1^{1/\delta}] = E[(X_0 + Z_1)^{1/\delta}] = E[X + Z] = x + E[Z]$
 So, $E_x[g(X)] - g(x) = E[Z] + x - x^{1/\delta} \rightarrow -\infty$ as $x \rightarrow \infty$
 (choose x_0 s.t. $E[Z] + x_0 - x_0^{1/\delta} = -1$)

Note that by Regenerative SLLN,
 $\frac{1}{n} \sum_{j=0}^{n-1} X_j \xrightarrow{a.s.} \frac{E_0[\sum_{j=0}^{n-1} X_j]}{E_0[\tau_1]} = \frac{E_0[\sum_{j=0}^{n-1} X_j]}{E_0[\tau_1]} = \frac{E_0[\frac{\tau_1(\tau_1-1)}{2}]}{E_0[\tau_1]}$
 And by Regenerative CLT,

$\sup_{x \leq x_0} E_x[g(X)] = \sup_{x \leq x_0} [x + E[Z]] = x_0 + E[Z] < \infty$
 $\inf_{x \leq x_0} E_x[g(X)] = \sup_{x \leq x_0} f_2(y^{1/\delta} - x)^{1/\delta-1} > 0$
 (pdf derivation)

3. Goal is to maximize $E_x[\sum_{j=0}^{\infty} r(X_j, A)]$ over $(A_n)_{n \geq 0}$ and $T = \inf \{n \geq 0 : X_n \in C\}$

4. $X_{n+1} = |X_n + Z_{n+1}|$, WTS $\pi(dx) = \frac{P(Z_1 > x)}{E[Z_1]} dx$

Construct MG:
 $M_n = \sum_{j=0}^n \gamma^j r(X_j, a^*(X_j)) + \mathbb{1}(T=n) \gamma^n r(X_n, a^*(X_n)) + \mathbb{1}(T > n) \gamma^n V^*(X_n)$ is MG
 Then $E_x[M_n - M_{n-1} | X_0 = x_0] = 0$, s.d.
 $\gamma^n r(X_{n-1}, a^*(X_{n-1})) \mathbb{1}(T \geq n) + \gamma^n \mathbb{1}(T > n) \gamma^n$
 reward from (n-1) timestep

$p(x, y) = \begin{cases} f_2(x+y), & 0 \leq x \leq y \\ f_2(x-y) + f_2(x+y), & x > y \end{cases}$
 Thus, $\int_0^\infty \pi(dx) p(x, y) dy = \pi(dy)$
 expected reward after time $n \leq 0$

Thus,
 $V^*(x) = \max_{a \in A} \left[r(x, a) + \gamma \sum_{y \in C} P_a(x, y) \max_{a \in A} r(y, a) + \gamma \sum_{y \in C} P_a(x, y) V^*(y) \right]$
 is the opt. eqn, and $V^*(x) \geq E_x[\sum_{j=0}^{\infty} \gamma^j r(X_j, A)]$

if we don't take optimal action (super MG case)
 assuming bounded rewards

Assignment 3.1:
 $X = (X_n)_{n \geq 0}$ is reducible, positive recurrent MC on discrete state space S . $f \in L^1(\pi) = \{f : \sum_x \pi(x) |f(x)| < \infty\}$
 fix z to be the "regeneration state" for non-type h
 $\pi h > \frac{E_x[\sum_{j=0}^{n(z)-1} h(X_j)]}{E_z[\tau(z)]}$ Thus $E_x[\sum_{j=0}^{n(z)-1} |f(X_j)|] < \infty$

Change of Measure
 $P(A), L \geq 0, E_P[L] = 1$
 $Q(dw) = L(w) P(dw)$ prob. on Ω
 $E_Q[Z] = E_P[Z] = E_P[ZL]$
 $\rightarrow E_P[Z] = E_Q[ZL^{-1}]$ "importance sampling"